Math 170E Notes - W3

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1 W3 (Ch1.3-1.5)

Example 1.1. How many 14 letter words we can create (meaningful or not) using the letters of "UCLAASHECENTER"?

(A)14! (B) $\frac{14!}{2}$ (C) $\frac{14!}{12}$ (D) $\frac{14!}{24}$ (E) $\frac{14!}{120}$

Solution

Remark 1.1 (Permutation of Letters)**.** In general, if we have n objects in total, among which n_i of them are the same for $i = 1, 2, \ldots, k$, there are

$$\binom{n}{n_1, n_2, \ldots, n_k} = \frac{n!}{n_1! n_2! \cdots n_k!}$$

total ways to permute them. Here of course $n_1 + n_2 + \cdots + n_k = n$.

Conditional Probability

Consider the first example from the first discussion, that we have a class of 60 as shown in the table

Class		
	Boys	Girls
Blue	10	15
Green	25	10

The teacher chooses one student uniform at random in the class but does not tell the students. She wants the students to guess the eye color of S. At first, betting that S has green eyes makes more sense as there are more 35 students with green eyes compared to the other 25 students with blue eyes. In probabilistic notation, $P(A) = \frac{35}{60} > \frac{1}{2}$ where $A = \{S \text{ has green eyes}\}$. Then, the teacher gives a hint that S is a girl. Now this changes everything, because we have more information because we can restrict the set of possibilities of S.

Now we need look at $\mathbb{P}(A|B)$ where $B = \{S \text{ is a girl}\}$.

Say that student is S. Let's define events $A = \{S \text{ is a boy}\}$, $B = \{S \text{ has green eyes}\}$. The sets A and B can be represented as yellow and red colored areas

	Boys	Girls
Blue	10	15
Green	25	10
The set B		

In this restriction, S is more likely to have blue eyes, with 15 to 10 favor. So we write $\mathbb{P}(S \text{ has blue eyes}|B) = \frac{15}{25}$. Define $C = \{S \text{ has blue eyes}\}$, we can also compute this as

$$\mathbb{P}(C|B) = \frac{\mathbb{P}(C \cap B)}{\mathbb{P}(B)} = \frac{\frac{15}{60}}{\frac{25}{60}} = \frac{15}{25}$$

Note that $\mathbb{P}(C|B)$ and $\mathbb{P}(B|C)$ are completely different things.

Example 1.2. Monica throws two dice in a backgammon game. You know that the sum of two dice is 10. What is the probability that the first dice is 5?

Solution

Independence

Now consider a different class with the same game.

	Boys	Girls
Blue	10	20
Green	20	40

Now compute $P(C)$ and $P(C|B)$. Are they different? Nope, you will find both equal to $\frac{2}{3}$, which means that the information B is not useful for us, i.e. knowing event B happens or not does not say anything about event C. This is exactly what *independence* is. If we have $P(C) = P(C|B)$, we say the events B and C are independent.

Example 1.3. (PSI 1.3.7) Rachel throws two dice in a backgammon game. You know that the sum of two dice is 7. What is the probability that the first dice is 5?

Solution

Ch1.5: Bayes's Theorem Examples

Example 1.4. (PSI 1.5.9 - modified) . There is a new diagnostic test for *coronavirus* that occurs in about 10% of the population. The test is not perfect, but will detect a person with the disease 95% of the time. It will, however, say that a person without the disease has the disease about 20% of the time. A person is selected at random from the population, and the test indicates that this person has the disease. What is the conditional probabilities that the person actually has the disease?

(A) $\frac{1}{3}$ (B) $\frac{9}{100}$ (C) $\frac{19}{55}$ (D) $\frac{15}{66}$ (E) $\frac{33}{95}$

Solution